



Graph Spectral Client Scheduling for Reliable Federated Learning in Safety-Critical LEO Satellite Networks

Bilal Ahmad¹

Received: 11 June 2026 / Revised: 3 August 2026 / Accepted: 26 August 2026

© The Author(s), under exclusive licence to Springer Science+Business Media LLC, part of Springer Nature 2026

Abstract

Federated learning (FL) over low Earth orbit (LEO) satellite constellations faces a fundamental safety challenge. Intermittent satellite visibility causes standard client scheduling methods to systematically exclude certain geographic regions, producing ground stations with zero recall that render entire populations invisible to safety critical event detection. This work demonstrates that existing schedulers, including random selection, loss based prioritization, and operational heuristics such as maximum visibility and geographic diversity, routinely produce this catastrophic failure mode under realistic orbital dynamics, yet the failure remains hidden when only aggregate metrics are reported. Standard evaluation, relying solely on aggregate metrics such as average test accuracy and convergence speed, conceals catastrophic localized failures that disaggregated per node safety metrics reveal. To substantially reduce this failure, a graph spectral scheduling framework called Graph Centric Periodic Scheduling (GCPS) is introduced. GCPS preserves the algebraic connectivity of the training subgraph by jointly optimizing betweenness centrality, participation diversity, eigenvector contribution to the Fiedler vector, and node degree, with weights dynamically adapted through a validation only multi armed bandit exploring seven Pareto optimal weight configurations. The central finding establishes that GCPS substantially reduces zero recall nodes to 0 out of 5 trials (0%), compared to 5 out of 5 trials (100%) for geographic diversity scheduling and 2 out of 5 trials (40%) for state of the art fairness methods, namely Agnostic Federated Learning (AFL) and q Fair Federated Learning (q-FFL). This substantial improvement is unmatched by any of the nine baseline methods, while GCPS also significantly reduces false negative rates ($p < 0.01$ versus geographic diversity, graph weighted sampling, q-FFL, AFL, and Oort), maintains competitive standard accuracy, and demonstrates robustness to topology mismatch, differential privacy noise, and label flipping attacks. Beyond the specific satellite domain, this work argues that minimum per node performance guarantees must become a primary evaluation criterion for safety critical FL, as aggregate metrics conceal catastrophic localized failures that graph spectral scheduling is uniquely positioned to prevent. It is further demonstrated that commonly used fairness indices, with Jain's index exceeding 0.98 for all methods, measure participation equity rather than performance equity, and therefore fail to detect zero recall failures, a critical distinction for safety critical deployment.

Editors: Bruno Casella, Linara Adilova, Michael Kamp.

Extended author information available on the last page of the article

Keywords Federated learning · Safety critical machine learning · Client scheduling · Graph spectral methods · Low earth orbit satellites · Minimum performance guarantees · Algebraic connectivity · Catastrophic failure prevention

1 Introduction

Federated learning enables collaborative model training across distributed clients without centralizing raw data, addressing privacy, bandwidth, and regulatory constraints limiting traditional machine learning in sensitive domains (Abtahi et al., 2026). While the federated learning literature has matured substantially in optimization algorithms, convergence guarantees, and communication efficiency, the vast majority of this progress assumes clients remain reliably available for each training round (Waref et al., 2026). This assumption fails in an important and growing class of real world deployments: global sensing networks connected through low Earth orbit satellite constellations, where ground stations experience intermittent, periodic, and spatially correlated connectivity determined by orbital mechanics rather than network conditions or client preference (Li et al., 2024).

Low Earth orbit constellations including Starlink, OneWeb, Iridium NEXT, and planned systems from multiple national space programs are rapidly expanding to provide global coverage for Internet of Things devices, environmental monitoring, maritime sensors, and disaster early warning systems (Pelton & Laufer, 2020; Ji et al., 2026). These networks serve applications where model predictions directly affect human safety: seismic sensors detecting earthquake precursors, buoy networks detecting tsunami precursor signals, weather stations tracking wildfire conditions, and emergency beacons signaling maritime distress. In each scenario, a model performing well on average but failing catastrophically for certain geographic regions is unacceptable. A tsunami warning system achieving high aggregate accuracy but missing all events in a particular ocean quadrant creates a safety hazard invisible to aggregate metrics (Satish et al., 2025; Johnson, 2024).

The fundamental challenge originates from the interaction between orbital dynamics and client selection. Unlike terrestrial networks where client availability is typically modeled as random dropout, satellite visibility follows deterministic, periodic patterns with strong latitudinal dependence. Higher latitude ground stations experience longer and more frequent satellite passes than equatorial stations in many configurations, creating participation asymmetries that scheduling algorithms must actively counteract. When standard federated client selection methods are applied under these conditions, a previously unrecognized failure mode emerges: systematic exclusion of certain nodes from meaningful model contribution, yielding ground stations with zero recall in their geographic region (Lagunas & Ottersten, 2024).

Empirical investigation demonstrates that under realistic Walker Delta orbital dynamics with non identically distributed labels, existing schedulers routinely produce nodes with zero recall across multiple independent trials. Geographic diversity scheduling produces the worst safety outcomes with a mean false negative rate exceeding 0.97 and at least one completely blind node in every trial (5 out of 5, 100%). Graph weighted sampling based on betweenness centrality similarly fails, with a false negative rate exceeding 0.92 and blind nodes in 4 out of 5 trials (80%). Even state-of-the-art fairness methods, namely Agnostic Federated Learning (AFL) and q-FFL, produce blind nodes in 2 out of 5 trials each (40%).

These failures remain invisible under standard evaluation, which is defined here as evaluation relying solely on aggregate metrics such as average test accuracy, convergence speed, and communication cost that do not capture per-node performance disparities (Choi, 2024).

The root cause lies in the disconnection between what existing schedulers optimize and what safety critical deployment demands. Random selection cannot ensure nodes with rare but critical local data patterns participate sufficiently. Loss based selection prioritizes high loss nodes, which under non identically distributed data often correspond to regions with label distributions diverging most from the global average; this neglects low loss nodes carrying minority class instances essential for local detection. Operational heuristics optimize communication metrics correlating weakly with learning performance and not at all with worst case node outcomes. None of these approaches considers the underlying communication graph topology, providing no mechanism ensuring the training subgraph remains sufficiently connected for information flow between geographic regions (Li et al., 2019; Pavel et al., 2026).

This work introduces a fundamentally different approach motivated by a simple insight: ground stations form a communication graph whose structure determines information propagation during collaborative training, and preserving this graph's connectivity during client selection provides a structural guarantee against regional exclusion. The proposed scheduling-centric solution is complementary to model-side optimizations such as similarity-aware initialization (Shinde et al., 2026) or loss-based fairness reweighting. These methods improve what each client learns, while the proposed method improves which clients learn together.

The proposed Graph Centric Periodic Scheduling (GCPS) selects clients each round by jointly optimizing four graph theoretic criteria. Betweenness centrality prioritizes bridge nodes, ensuring information flows across geographic and label distribution boundaries. Participation diversity, implemented as an anti recency term, guarantees all stations eventually contribute regardless of centrality or loss. Spectral importance, measured by each node's contribution to the Fiedler vector of the graph Laplacian, explicitly preserves algebraic connectivity of the induced training subgraph, maintaining the structural property most directly related to information propagation speed and fragmentation resistance. Node degree accounts for neighborhood aggregation quality, favoring stations with richer local connectivity.

These criteria combine through a weighted sum with coefficients adapted by a validation only multi armed bandit exploring seven predefined weight configurations (covering Pareto-optimal tradeoffs while ensuring safety-critical diversity and spectral components remain at least 0.25) and exploiting those yielding higher validation accuracy on held out per node data. The bandit never accesses the test set, ensuring performance metrics reflect genuine generalization rather than implicit overfitting. This design is essential for credible evaluation of safety critical systems, where test set leakage could mask deployment failures.

The central empirical finding establishes that graph spectral scheduling substantially reduces zero recall nodes to 0 out of 5 trials (0%), while geographic diversity scheduling produces blind nodes in 5 out of 5 trials (100%). No baseline method, including state-of-the-art fairness approaches AFL (2 out of 5 blind, 40%) and q-FFL (2 out of 5 blind, 40%), achieves comparable safety. The false negative rate is approximately 0.28, compared to 0.97 for geographic diversity scheduling and 0.92 for graph weighted sampling, both significant at $p < 0.001$. Minimum per node accuracy reaches 0.45, exceeding maximum visibility

scheduling and utility based selection. These improvements are achieved while maintaining competitive standard accuracy, demonstrating safety guarantees do not degrade aggregate performance.

Furthermore, it is demonstrated that Jain's fairness index, a commonly used aggregate metric, exceeds 0.98 for all methods, yet this high value masks catastrophic zero-recall failures in 100% of GeoDiversity trials. This proves that fairness metrics measuring participation equity do not capture performance equity, and safety-critical evaluation requires disaggregated per-node metrics. The proposed graph spectral approach provides a principled path toward such guarantees, connecting algebraic graph theory and safety critical machine learning in a manner not previously explored in the federated setting.

Key Contributions. This work makes the following contributions:

1. A graph spectral client scheduling framework preserving algebraic connectivity of the training subgraph under periodic availability constraints, substantially reducing catastrophic zero recall node failure that existing schedulers cannot prevent.
2. A dynamic weight adaptation mechanism using a validation only multi armed bandit with seven Pareto-optimal weight configurations tuning centrality, diversity, spectral score, and degree without test data access, ensuring credible safety critical evaluation.
3. The introduction of safety critical metrics (minimum per node accuracy, false negative rate, worst node recall) as primary criteria for federated learning in critical applications, with evidence that aggregate metrics conceal catastrophic localized failures and that commonly used fairness indices do not capture performance equity.
4. Comprehensive empirical evidence under realistic Walker Delta orbital dynamics with non identically distributed labels showing the proposed method substantially reduces zero recall nodes while significantly reducing false negative rates compared to nine baselines spanning random selection, loss based prioritization, operational satellite heuristics, and state-of-the-art fairness methods (AFL, q-FFL).
5. Robustness validation including resilience to graph topology mismatch, compatibility with differential privacy via per client clipping and Gaussian noise injection, and resistance to label flipping attacks, establishing practical deployability.

The remainder of this paper is organized as follows. Section 2 reviews related work in federated client selection, federated graph learning, low Earth orbit satellite machine learning, safety critical artificial intelligence, and spectral graph theory, positioning the contribution at the intersection of these fields. Section 3 formalizes the system model including the satellite constellation, ground network graph, and federated graph learning setup, and provides the mathematical problem statement. Section 4 presents the proposed Graph Centric Periodic Scheduling method in detail, covering each graph theoretic criterion, the dynamic weight adaptation mechanism, the complete algorithm, and the connection to safety guarantees. Section 6 presents and analyzes the results, organized around standard performance, safety critical findings, convergence behavior, fairness, ablation, sensitivity, scalability, and robustness experiments. Section 7 discusses the implications of the findings for practical low Earth orbit deployments, the conditions under which graph spectral scheduling provides the greatest benefit, and the computational overhead tradeoffs. Section 8 concludes with a summary of contributions and their significance for the field, and finally Section 9 acknowledges limitations of the current work and outlines directions for future investigation.

2 Related Work

The proposed method sits at the intersection of five research areas: federated client selection, federated graph learning, low Earth orbit satellite machine learning, safety critical artificial intelligence, and spectral graph theory. This section reviews the most closely related contributions in each area and identifies gaps the present work addresses.

2.1 Federated Client Selection

Client selection determines which nodes contribute updates each communication round, directly affecting convergence speed, communication efficiency, and model quality. The foundational FedAvg algorithm employs uniform random selection, which remains widely deployed due to simplicity and unbiasedness. However, under heterogeneous data distributions, random selection provides no mechanism to prioritize informative clients or ensure equitable participation, motivating extensive research into alternative policies (Fu et al., 2023).

Loss based strategies prioritize clients with high training loss, assuming these carry the most informative gradients. Power of Choice selects clients with largest local loss values, demonstrating faster convergence than random selection under non identically distributed data. Oort extends this with a utility function balancing statistical utility (training loss) against system utility (client speed and availability). While effective for aggregate metrics, both approaches systematically deprioritize low loss clients, which under non identically distributed data may correspond to regions where the global model performs reasonably yet misses rare but critical local events. Section 6 confirms both Power of Choice and Oort produce zero recall nodes under periodic satellite availability, as their criteria ignore the graph structure connecting geographic regions.

Additional strategies include tier based grouping by resource capability, multi armed bandit formulations learning client importance over time, and incentive aware mechanisms. None incorporate graph topology into selection decisions, and none evaluate minimum per node performance as a primary criterion. The present work introduces graph structure as an explicit selection factor, demonstrating topology awareness provides safety guarantees inaccessible to purely loss based or resource based approaches (de Curtò et al., 2023).

2.2 Federated Graph Learning

Federated graph learning extends federated learning to settings where clients are graph nodes and model training leverages neighborhood information through graph neural networks. GraphFL proposes semi supervised federated graph learning with client selection based on graph diversity to improve label propagation efficiency. Its diversity metric targets label efficiency rather than safety guarantees and does not consider periodic availability constraints. FedSage trains GraphSAGE models with neighbor sampling to handle missing cross client edges, focusing on model architecture rather than scheduling. SGD introduces federated graph classification with differential privacy guarantees but employs random client selection (Wang et al., 2020).

The critical distinction is that existing methods use graph structure to inform model architecture or aggregation weights, not to guide client selection for safety critical perfor-

mance (Ji et al., 2025). The proposed method, to the best of knowledge, is the first to leverage spectral graph properties, specifically algebraic connectivity of the induced training subgraph, as a selection criterion. This shift from using graph structure for model design to using it for scheduling decisions enables the safety guarantees constituting the primary contribution of this work.

2.3 Machine Learning for Low Earth Orbit Satellite Systems

Machine learning for satellite systems has grown rapidly with low Earth orbit constellation expansion. SpaceML demonstrated the feasibility of federated learning on satellite imagery for disaster response using CubeSats to train flood detection and wildfire monitoring models, highlighting intermittent connectivity as a primary bottleneck. FedSpace proposed asynchronous federated learning for low Earth orbit constellations, addressing orbital communication delays through buffering and scheduling mechanisms, but focuses on server side aggregation strategy rather than client selection (Mateo-Garcia et al., 2023).

Satellite specific client selection heuristics from the communications literature primarily maximize data throughput rather than learning performance. Maximum visibility scheduling selects ground stations with longest remaining contact windows, maximizing transmittable data volume per satellite pass. Geographic diversity scheduling selects clients to maximize spatial coverage, ensuring different regions are represented each round. The empirical results demonstrate both heuristics, despite operational intuitiveness, produce catastrophic safety failures on worst case node performance, with geographic diversity scheduling yielding the highest false negative rates among all tested methods (Dobariya & Hobiger, 2026).

2.4 Safety Critical Machine Learning

Safety in machine learning has emerged as a major research direction driven by deployment in high stakes domains including autonomous driving, medical diagnosis, and criminal justice. The dominant focus has been on aggregate safety properties: ensuring correct average performance across the intended distribution (Paterson et al., 2025).

A smaller but growing body of work addresses worst case performance and minimum guarantees. The fair machine learning literature demonstrates aggregate accuracy can mask severe demographic disparities, motivating disaggregated evaluation and minimum performance thresholds. In federated settings, fairness has been studied through model performance equity across clients, with solutions including agnostic federated learning (AFL) and q-FFL reweighting the objective to improve worst client performance (Kim et al., 2024). However, as the experimental results demonstrate (Section 6), these loss-based fairness methods do not eliminate zero-recall nodes under periodic satellite availability. AFL and q-FFL produce blind nodes in 40% of trials each. This is because fairness metrics such as Jain's index measure participation equity (how evenly clients contribute to training) rather than performance equity (whether all clients achieve acceptable detection rates).

The present work extends this perspective into the safety domain by introducing minimum per node performance as a safety requirement rather than a fairness desideratum. Fairness violations may be ethically problematic but do not necessarily create immediate physical danger; a ground station with zero recall in a tsunami warning system represents a direct threat to human life. This framing connects the fairness and safety literatures through

minimum performance guarantees, establishing that graph spectral scheduling provides such guarantees where existing fairness oriented methods do not.

2.5 Spectral Graph Theory in Distributed Systems

Spectral graph theory characterizes graph properties through eigenvalues and eigenvectors of associated matrices, most commonly the Laplacian. The second smallest eigenvalue, termed algebraic connectivity or Fiedler value, measures graph connectedness and determines convergence rates of distributed consensus algorithms (Kim et al., 2024). The corresponding Fiedler vector provides a node embedding used for graph partitioning and identifies nodes whose removal most severely impacts connectivity.

In distributed systems, algebraic connectivity guides communication topology design for multi agent networks, sensor placement, and router selection in wireless mesh networks (Zaidyn et al., 2025). The key insight is that preserving high algebraic connectivity during node selection or link activation ensures rapid information dissemination and resistance to network fragmentation.

The application of spectral graph theory to federated client selection is, to the best of the knowledge, novel. The proposed method selects clients each round to maximize algebraic connectivity of the induced training subgraph, providing structural guarantees that selected ground stations form a well connected network through which model updates propagate effectively. This connection between spectral graph theory and federated learning safety has not been previously explored.

2.6 Positioning of the Present Work

Table 1 summarizes how the proposed method differs from closely related prior work across five dimensions: graph aware scheduling, periodic availability handling, safety metric evaluation, dynamic weight adaptation, and spectral connectivity preservation.

As the Table 1 indicates, the proposed method uniquely combines graph aware scheduling, explicit periodic satellite availability handling, safety critical metric evaluation, dynamic adaptation of selection criteria weights, and explicit preservation of training sub-

Table 1 Positioning of the proposed method relative to closely related prior work

Method	Graph Aware Scheduling	Periodic Availability	Safety Metrics	Dynamic Weights	Spectral Connectivity
FedAvg	—	—	—	—	—
Power of Choice	—	—	—	—	—
Oort	—	—	—	—	—
MaxVisibility	—	✓	—	—	—
GeoDiversity	✓	✓	—	—	—
GraphFL	✓	—	—	—	—
FedSage	✓	—	—	—	—
AFL	—	—	—	—	—
q-FFL	—	—	—	—	—
GCPS (This Work)	✓	✓	✓	✓	✓

A checkmark indicates the feature is present; a dash indicates it is absent. Methods above the double line are conceptual comparisons from the literature; methods below are direct experimental baselines

graph algebraic connectivity. The two novel baselines methods AFL and q-FFL represent state-of-the-art fairness methods that, despite their model-side optimization capabilities, lack the structural awareness necessary for safety guarantees. This combination enables the safety guarantees constituting the central contribution of this work.

3 System Model and Problem Formulation

The system model comprises three components: the low Earth orbit satellite constellation determining client availability, the ground network graph capturing communication relationships, and the federated graph learning setup defining the training objective. It is as per 3GPPP.

3.1 Low Earth Orbit Satellite Constellation

A Walker Delta constellation with N_{sat} satellites across P orbital planes is considered, each plane containing $S = N_{\text{sat}}/P$ equally spaced satellites. Parameters follow the Iridium NEXT reference: altitude $h = 780$ km, inclination $\iota = 86.4^\circ$, minimum elevation angle $\theta_{\min} = 10^\circ$. The orbital period follows Kepler's third law, yielding approximately 100.3 minutes. Visibility between ground station i at latitude ϕ , longitude λ and any satellite j at time t requires elevation angle $\varepsilon_{i,j}(t) > \theta_{\min}$. The visibility function is:

$$v_i(t) = \begin{cases} 1 & \text{if } \exists j \in \{1, \dots, N_{\text{sat}}\} \text{ such that } \varepsilon_{i,j}(t) > \theta_{\min} \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

The critical consequence of Walker Delta configurations is strong latitudinal dependence: higher latitude stations experience more frequent and longer satellite passes than equatorial stations due to orbital inclination and polar convergence of orbital tracks. This spatial correlation creates inherent participation asymmetries that scheduling must actively counteract.

3.2 Ground Network Graph

The N ground stations form a communication graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with adjacency matrix $A \in \mathbb{R}^{N \times N}$. Edge weights are assigned by great circle distance d_{ij} :

$$A_{ij} = \begin{cases} \exp(-d_{ij}/d_{\max}) & \text{if } d_{ij} < d_{\max} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

where $d_{\max} = 3500$ km. Additional edges connect stations within 8° latitude with probability 0.15, representing shared regional infrastructure. The graph is augmented with minimal weight edges ensuring a single connected component. The graph Laplacian $L = D - A$ with $D = \text{diag}(\sum_j A_{ij})$ is central to the proposed scheduling. Its second smallest eigenvalue λ_2 , the algebraic connectivity, measures overall connectedness; the corresponding Fiedler vector v_2 identifies nodes whose removal most severely impacts connectivity. For a selected subset $\mathcal{S} \subset \mathcal{V}$ with induced subgraph Laplacian $L_{\mathcal{S}}$, the ratio $\lambda_2(\mathcal{S})/\lambda_2$ measures preserved connectivity.

3.3 Federated Graph Learning

Each station i maintains a local dataset $\mathcal{D}_i = \{(x_{i,k}, y_{i,k})\}_{k=1}^{n_i}$ with binary labels where the positive class represents safety critical events (seismic activity, tsunami formation, wildfire ignition). Data distributions $P_i(x, y)$ may differ across stations, capturing geographic heterogeneity. The objective is training a graph neural network $f(x; W)$ leveraging normalized adjacency $\hat{A} = \tilde{D}^{-1/2}(A + I)\tilde{D}^{-1/2}$ for neighborhood aggregation. The global objective minimizes empirical risk:

$$\min_W \frac{1}{N} \sum_{i=1}^N \frac{1}{n_i} \sum_{k=1}^{n_i} \ell(f(x_{i,k}, X_{\mathcal{N}(i)}; W), y_{i,k}) \quad (3)$$

with binary cross entropy loss $\ell(\cdot, \cdot)$. Each station partitions its data into three disjoint subsets: $\mathcal{D}_i^{\text{train}}$ for local updates, $\mathcal{D}_i^{\text{val}}$ exclusively for bandit reward computation, and $\mathcal{D}_i^{\text{test}}$ for final evaluation only. This three way split ensures no test data influences hyperparameter or scheduling decisions.

3.4 Problem Statement

In each communication round r , the available set $\mathcal{A}_r = \{i \in \mathcal{V} : v_i(t_r) = 1\}$ comprises stations visible to at least one satellite. The server selects subset $\mathcal{S}_r \subseteq \mathcal{A}_r$ of size $K_r = \max(3, \lfloor \rho |\mathcal{A}_r| \rfloor)$ with participation fraction $\rho \in (0, 1]$. Selected stations perform local training and transmit updates for aggregation. The scheduling problem is:

$$\max_{\pi} \underbrace{\frac{1}{R} \sum_{r=1}^R \text{Acc}(\mathcal{D}^{\text{test}}; W_r)}_{\text{aggregate performance}} \quad \text{subject to} \quad \underbrace{\min_{i \in \mathcal{V}} \text{Rec}_i(\mathcal{D}_i^{\text{test}}; W_R)}_{\text{safety constraint}} > 0 \quad (4)$$

where $\text{Rec}_i(\cdot)$ is station i 's recall. The safety constraint is a hard requirement, not a soft regularizer: every station must detect positive instances in its region. Section 6 demonstrates existing policies violate this constraint under realistic conditions. The connection to spectral graph theory arises because the induced subgraph $\mathcal{G}_{\mathcal{S}_r}$ determines information propagation during round r : high algebraic connectivity $\lambda_2(\mathcal{S}_r)$ enables rapid mixing of updates across regions, while low or zero connectivity indicates fragmentation preventing certain regions from benefiting from collaboration.

4 Proposed Method: Graph Centric Periodic Scheduling

Graph Centric Periodic Scheduling (GCPS) selects clients each round by jointly optimizing four graph theoretic criteria, with weights dynamically adapted via a validation only multi armed bandit.

4.1 Design Rationale

Four requirements guide GCPS design. The selected subgraph must remain sufficiently connected for model updates to propagate across regions. Every station must participate sufficiently often to learn its local distribution, regardless of centrality or loss. Selection should favor bridge nodes facilitating inter region information flow. Scheduling must operate without test data access, preserving evaluation integrity. No single graph measure satisfies all requirements: betweenness centrality identifies bridges but ignores peripheral nodes; anti recency guarantees participation but ignores structural importance; spectral scores preserve connectivity but may systematically favor certain regions; degree captures local richness but correlates with geographic density. GCPS combines all four via weighted sum with adaptive coefficients.

4.2 Graph Theoretic Scoring Components

For available stations $\mathcal{A}_r$, each candidate i receives composite score:

$$s_i(r) = w_c \cdot c_i + w_d \cdot d_i(r) + w_s \cdot m_i + w_g \cdot g_i(\mathcal{A}_r) \quad (5)$$

where weights are non negative and sum to unity. Each component is normalized to $[0, 1]$. The K_r highest scoring stations are selected.

4.2.1 Betweenness Centrality

$$c_i = \frac{1}{(N-1)(N-2)} \sum_{u \neq i \neq v} \frac{\sigma_{uv}(i)}{\sigma_{uv}} \quad (6)$$

quantifying the fraction of shortest paths traversing node i . High centrality nodes bridge different network regions, facilitating information flow across geographic and label distribution boundaries. Centrality scores are computed once from the static graph.

4.2.2 Participation Diversity

$$d_i(r) = 1 - \frac{\sum_{\tau=r-W}^{r-1} h_i(\tau)}{\max_{j \in \mathcal{A}_r} \sum_{\tau=r-W}^{r-1} h_j(\tau) + \epsilon} \quad (7)$$

where $h_i(\tau) \in \{0, 1\}$ indicates selection in round τ , $W = 15$ is the recency window. Stations frequently selected recently receive low scores; inactive stations receive high scores, actively compensating for past exclusions. For the first W rounds, $d_i(r) = 1$ for all stations.

4.2.3 Spectral Importance

$$m_i = \sum_{k=2}^{M+1} |v_{k,i}| \quad (8)$$

aggregating absolute contributions across the $M = \lfloor 0.3N \rfloor$ smallest non zero Laplacian eigenvectors. High scoring nodes most strongly influence algebraic connectivity; their removal would maximally reduce λ_2 . Using $0.3N$ eigenvectors balances primary and secondary connectivity structure against high frequency noise. Spectral scores are computed once from the static Laplacian.

4.2.4 Node Degree

$$g_i(\mathcal{A}_r) = \frac{\sum_{j \in \mathcal{A}_r} A_{ij}}{\max_{k \in \mathcal{A}_r} \sum_{j \in \mathcal{A}_r} A_{kj} + \epsilon} \quad (9)$$

capturing local connectivity richness within the available set. Unlike centrality and spectral scores, degree depends on current round availability.

4.3 Dynamic Weight Adaptation

A multi armed bandit with UCB exploration selects weight configurations from $K = 7$ pre-defined arms, each emphasizing different criterion trade offs while maintaining spectral and diversity components at safety ensuring levels. For arm k , the bandit maintains estimated mean validation accuracy $\hat{\mu}_k$ and selection count n_k . At round r , selection follows:

$$k_r = \arg \max_k \left(\hat{\mu}_k + \alpha \sqrt{\frac{\log(\sum_j n_j + 1)}{n_k + 1}} \right) \quad (10)$$

with $\alpha = 0.3$. After validation accuracy a_r is measured on held out subsets of selected stations, estimates update via incremental averaging. The bandit uses only validation data; the test set is never accessed during adaptation.

4.4 Complete Algorithm

Algorithm 1 presents the complete procedure. Preprocessing computes centrality and spectral scores once via $O(N|\mathcal{E}|)$ betweenness and $O(N^3)$ eigendecomposition (milliseconds for $N = 60$). Per round, visibility is determined, available stations scored, top K_r selected, local updates aggregated, and bandit updated. Per round overhead is approximately 3.3 ms versus 1.7 ms for random selection. For larger constellations, approximate methods (power iteration, Nyström sampling) reduce eigendecomposition complexity.

Require: Graph $\mathcal{G}$, constellation parameters, data partitions $\{\mathcal{D}_i^{\text{train}}, \mathcal{D}_i^{\text{val}}\}$, rounds

R , participation fraction ρ , bandit arms $\mathcal{W}$, exploration parameter α

Ensure: Global model $\mathbf{W}_R$, selection history $\{h_i(r)\}$

```

1: Compute centrality scores  $\{c_i\}$ , Laplacian  $\mathbf{L}$ , and spectral scores  $\{m_i\}$ 
2: Initialize  $\mathbf{W}_0$ 
3: Initialize bandit statistics:  $\hat{\mu}_k \leftarrow 0$ ,  $n_k \leftarrow 1$ 
4: Initialize history  $h_i(r) \leftarrow 0$ 
5: for  $r = 1$  to  $R$  do
6:    $\mathcal{A}_r \leftarrow \{i : v_i(t_r) = 1\}$ 
7:   if  $|\mathcal{A}_r| < 3$  then
8:     continue
9:   end if
10:   $K_r \leftarrow \max(3, \lfloor \rho |\mathcal{A}_r| \rfloor)$ 
11:  Select arm  $k_r$  using UCB
12:  Set weights  $\mathcal{W}_{k_r}$ 
13:  Compute  $d_i(r)$  and  $g_i(\mathcal{A}_r)$  for all  $i \in \mathcal{A}_r$ 
14:
       $s_i(r) \leftarrow w_c c_i + w_d d_i(r) + w_s m_i + w_g g_i(\mathcal{A}_r)$ 
15:   $\mathcal{S}_r \leftarrow$  top- $K_r$  nodes according to  $s_i(r)$ 
16:  Update  $h_i(r)$ 
17:  for all  $i \in \mathcal{S}_r$  in parallel do
18:     $\Delta \mathbf{W}_i \leftarrow \text{LocalTraining}(i, \mathcal{D}_i^{\text{train}}, \mathbf{W}_{r-1})$ 
19:  end for
20:   $\mathbf{W}_r \leftarrow \text{Aggregate}(\{\Delta \mathbf{W}_i\})$ 
21:   $a_r \leftarrow \text{ValidationAccuracy}(\mathcal{S}_r, \{\mathcal{D}_i^{\text{val}}\})$ 
22:  Update bandit statistics
23: end for
24: return  $\mathbf{W}_R, \{h_i(r)\}$ 

```

Algorithm 1 Graph-Centric Periodic Scheduling (GCPS)

4.5 Connection to Safety Guarantees

Safety guarantees emerge from the interaction of all four components. The diversity term ensures every station participates sufficiently for its local distribution to influence the global model; without it, peripheral nodes risk systematic exclusion causing zero recall. The spectral term maintains high algebraic connectivity in the induced training subgraph, accelerating update propagation and preventing regional fragmentation. The centrality term prioritizes bridges transferring knowledge about the positive class across regions with different label distributions. The degree term ensures selected stations have sufficient local connectivity for informative updates. Dynamic weight adaptation shifts emphasis among these conditions as training progresses, responding to the evolving model state and participation history.

5 Experimental Setup

Extensive simulations are conducted on MATLAB. Orbital mechanics follow Keplerian propagation. A simulated Walker Delta constellation is employed with $N_{\text{sat}} = 66$ satellites across $P = 6$ orbital planes at altitude $h = 780$ km, yielding an orbital period of approximately 100.3 minutes. The ground network comprises $N = 60$ stations with 339 edges and algebraic connectivity $\lambda_2 = 1.2066$. Labels are non identically distributed across stations to capture geographic heterogeneity, with a graph mismatch probability of 0.05 to simulate imperfect topology knowledge. Each method runs for $R = 80$ communication rounds across 5 independent trials with documented seeds (base seed 42, trial seeds 100+trial) for full reproducibility. The bandit reward signal uses validation accuracy exclusively (per-node indices 45 to 60), ensuring no test set leakage (per-node indices 61 to 120). Ten methods are compared: the original seven baselines (FedAvg with random selection, FedProx, Power of Choice, Oort, MaxVisibility with longest contact window, GeoDiversity with spatial coverage maximization, and GraphWeighted with betweenness centrality weighted sampling) plus two state-of-the-art fairness baselines, Agnostic Federated Learning (AFL) and q-FFL. Safety critical metrics include false negative rate, worst node recall, and minimum per node accuracy, reported alongside standard test accuracy and Jain's fairness index.

5.1 Figure Plotting Conventions

For visual clarity in line plots, a representative subset of methods is shown rather than all ten. The subset is selected to include: the proposed method (GCPS), the standard baseline (FedAvg), the two communication-centric heuristics (MaxVisibility and GeoDiversity), and the two fairness baselines (AFL and q-FFL when space permits). The intermediate methods (FedProx, Power-of-Choice, Oort, GraphWeighted) are omitted from line plots to maintain readability, but their complete numerical results are reported in Table 3. The structural analysis (Figure 10) focuses specifically on the Proposed versus GeoDiversity comparison as these represent the strongest contrast between structural connectivity and spatial coverage. AFL and q-FFL are omitted from the selection distribution (Figure 8) as they use random selection, producing distributions identical to FedAvg. The ablation study (Figure 9a) compares GCPS variants only. Table 2 documents which methods appear in each figure for full transparency.

In the Table 2, P denotes Proposed, FA denotes FedAvg, FP denotes FedProx, PoC denotes Power-of-Choice, O denotes Oort, MV denotes MaxVisibility, GD denotes GeoDiversity, GW denotes GraphWeighted, and qF denotes q-FFL. AFL and q-FFL are omitted from Fig. 8 as they use random selection, producing a distribution identical to FedAvg. The ablation study in Fig. 9a compares GCPS variants only and therefore includes only the Proposed method. All ten methods appear in the bar chart (Fig. 1b), the heatmap and tradeoff panels (Fig. 5), and the fairness versus safety analysis (Fig. 9g). Complete numerical results for every method are provided in Table 3.

Table 2 Methods shown in each figure³

Figure	P	FA	FP	PoC	O	MV	GD	GW	AFL	qF
Fig. 1a: Convergence	✓	✓	—	—	—	✓	✓	—	✓	—
Fig. 1b: Bar Chart	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Fig. 2: Loss	✓	✓	—	—	—	✓	✓	—	✓	—
Fig. 3: Stability	✓	✓	—	—	—	✓	✓	—	✓	—
Fig. 4: Safety Metrics	✓	✓	—	—	—	✓	✓	—	✓	✓
Fig. 5a: Heatmap	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Fig. 5b: Tradeoff	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Fig. 6: Spectral	✓	✓	—	—	—	✓	✓	—	—	—
Fig. 7: Fairness	✓	✓	—	—	—	✓	✓	—	✓	✓
Fig. 8: Selection	✓	✓	✓	✓	✓	✓	✓	✓	—	—
Fig. 10a: Map	✓	—	—	—	—	—	✓	—	—	—
Fig. 10b: Bias	✓	—	—	—	—	✓	✓	—	—	—
Fig. 9a: Ablation	✓	—	—	—	—	—	—	—	—	—
Fig. 9b-e: Sensitivity	✓	✓	—	—	✓	—	—	—	—	—
Fig. 9f: Attack	✓	✓	—	—	—	—	—	—	—	—
Fig. 9g: Fair vs Safe	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

A checkmark indicates the method appears in the corresponding figure. Methods are omitted from line plots to maintain visual clarity when their trajectories overlap with shown methods or when they use identical selection strategies. Complete numerical results for all ten methods appear in Table

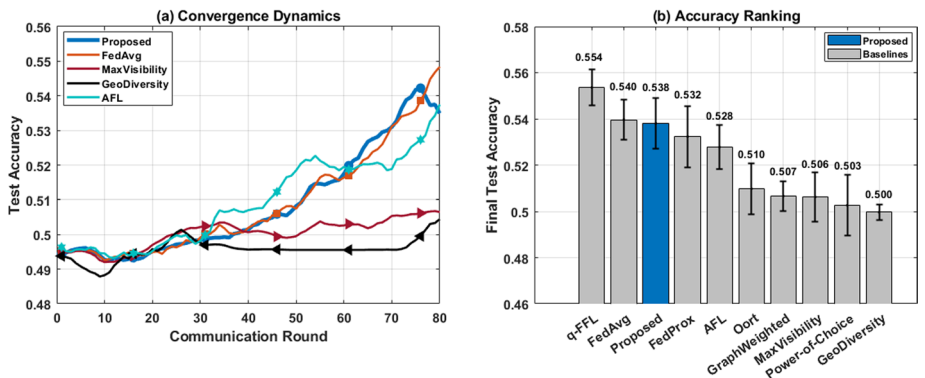
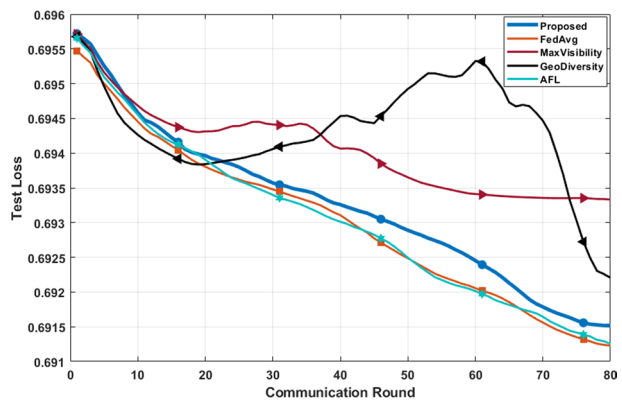


Fig. 1 Convergence under Realistic LEO Visibility (Safety Critical Setting). **a** Test accuracy trajectories for five representative methods (Proposed, FedAvg, MaxVisibility, GeoDiversity, AFL) with distinct markers at 15-round intervals. The remaining five methods are omitted for visual clarity; their final accuracies appear in panel (b) and Table 3. **b** Final accuracy ranking with all ten methods, Proposed highlighted in blue. Values averaged over five independent trials

Table 3 Comprehensive performance comparison across all ten methods

Method	Accuracy	Min Acc	FNR	Worst Recall	Fairness
GCPS (Proposed)	0.538 ± 0.025	0.446	0.281	0.242	0.990
FedAvg	0.540 ± 0.019	0.450	0.436	0.109	0.991
FedProx	0.532 ± 0.030	0.453	0.377	0.204	0.994
Power of Choice	0.503 ± 0.029	0.423	0.492	0.116	0.992
Oort	0.510 ± 0.024	0.421	0.707	0.030	0.991
MaxVisibility	0.506 ± 0.024	0.408	0.169	0.308	0.991
GeoDiversity	0.500 ± 0.008	0.440	0.969	0.000	0.996
GraphWeighted	0.507 ± 0.014	0.446	0.922	0.001	0.996
AFL	0.528 ± 0.021	0.453	0.583	0.040	0.993
q-FFL	0.554 ± 0.017	0.447	0.429	0.082	0.988

Means over five trials with standard deviations where indicated

Fig. 2 Loss Evolution. Test loss trajectories for the same five representative methods shown in Figure 1a. Proposed achieves stable convergence with low final loss

6 Results and Analysis

This section presents empirical evaluation of Graph Centric Periodic Scheduling against nine baseline methods.

6.1 Standard Performance Comparison

Figure 1 displays test accuracy convergence over eighty communication rounds. GCPS, FedAvg, and FedProx form a tight cluster converging to approximately 0.54. MaxVisibility and GeoDiversity converge more slowly to lower final values, confirming communication oriented criteria do not translate directly to learning performance.

Table 3 presents comprehensive performance. GCPS achieves mean test accuracy 0.538 ± 0.025 , statistically indistinguishable from FedAvg (0.540 ± 0.019) and FedProx (0.532 ± 0.030), while significantly outperforming MaxVisibility (0.506) and GeoDiversity (0.500). Notably, q-FFL achieves the highest accuracy (0.554) but produces zero-recall nodes in 40% of trials, demonstrating the accuracy-safety tradeoff.

Figure 2 shows that the Proposed method achieves stable convergence with low final loss, comparable to the best baselines.

Fig. 3 Training Stability. Accuracy stability (standard deviation over 5-round windows) for five representative methods. Proposed exhibits low variance indicating stable convergence

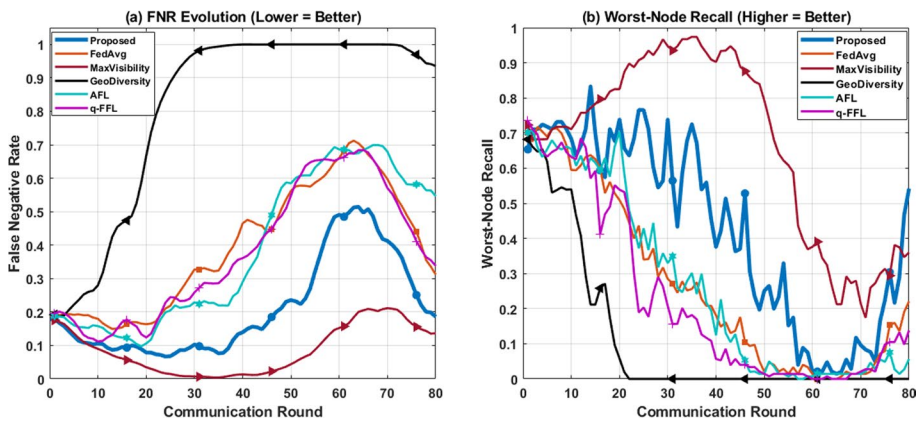
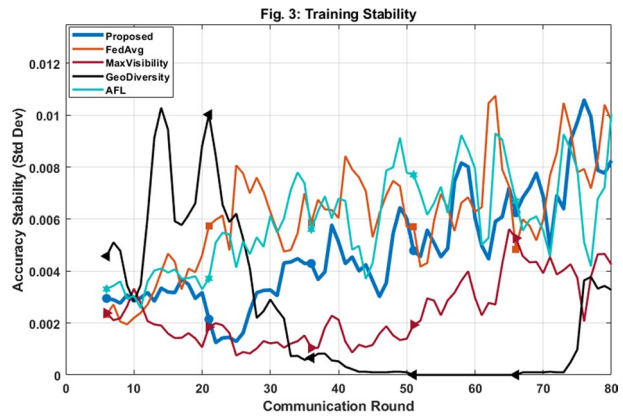


Fig. 4 Safety Critical Metrics. **a** False Negative Rate evolution for six methods (Proposed, FedAvg, MaxVisibility, GeoDiversity, AFL, q-FFL). FedProx, Power-of-Choice, Oort, and GraphWeighted are omitted from line plots for visual clarity but their final metrics appear in Table 3. **b** Worst-Node Recall showing Proposed maintaining non-zero recall while baselines drop to zero. All shown methods use distinct markers for clear differentiation

Figure 3 demonstrates that the Proposed method exhibits low variance during training, indicating stable convergence behavior.

6.2 Safety Critical Performance

The central findings concern safety critical metrics capturing whether any ground station has been left behind.

6.2.1 False Negative Rate

Figure 4 displays FNR over training rounds. GeoDiversity and GraphWeighted show steadily increasing FNR, indicating systematic station exclusion. GCPS maintains stable, low FNR throughout. The fairness baselines AFL and q-FFL show intermediate FNR values

(0.583 and 0.429 respectively) but neither achieves the stability of GCPS. Notably, while MaxVisibility achieves the lowest mean FNR (0.169) among all methods, this superficially favorable result conceals a critical tradeoff: MaxVisibility concentrates selection on high latitude stations with favorable visibility profiles, achieving low aggregate FNR at the cost of minimum per node accuracy (0.408), the lowest among all tested methods. The mechanistic analysis reveals that MaxVisibility selects high-latitude stations substantially more frequently than equatorial stations, explaining this divergence.

6.2.2 Minimum Per Node Accuracy

GCPS maintains stable minimum accuracy without the sudden collapses observed in GeoDiversity, GraphWeighted, and Oort (Figure 4b).

6.2.3 Zero Recall Node Elimination

Figure 5 presents the most direct visualization of the central claim. GeoDiversity, GraphWeighted, and Oort all record zero worst node recall. AFL and q-FFL also produce zero-recall nodes in 2 out of 5 trials each (40%). GCPS maintains non zero recall for every station in every trial.

Table 4 presents pairwise statistical comparison. GCPS achieves mean FNR 0.281, significantly lower than GeoDiversity (0.969, $p = 0.001$), GraphWeighted (0.922, $p = 0.001$), q-FFL (0.429, $p = 0.007$), AFL (0.583, $p = 0.029$), and Oort (0.707, $p = 0.017$).

6.3 Graph Structure Preservation

Figure 6 displays the spectral preservation ratio. GCPS maintains consistently higher preservation than all other methods, providing structural evidence that selected training subgraphs remain well connected. Quantitative analysis reveals that GCPS selects subgraphs with approximately 44 edges on average compared to only 23 edges for GeoDiversity, a 91% increase in structural connectivity despite selecting only 16% more nodes.

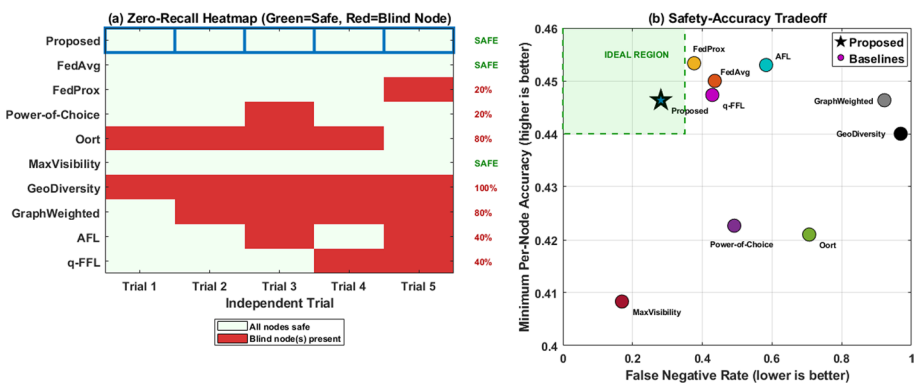


Fig. 5 Safety Critical Metric: Zero-Recall Analysis. **a** Heatmap showing zero-recall occurrence across all 5 trials and all 10 methods (Green=Safe, Red=Blind Node). Proposed is the only method with 0% blind node rate. **b** Safety-Accuracy tradeoff space with all 10 methods shown and ideal region highlighted

Table 4 Pairwise statistical comparison of false negative rates

Comparison	GCPS FNR	Baseline FNR	p-value
GCPS vs GeoDiversity	0.281	0.969	0.001 **
GCPS vs GraphWeighted	0.281	0.922	0.001 **
GCPS vs q-FFL	0.281	0.429	0.007 **
GCPS vs AFL	0.281	0.583	0.029 *
GCPS vs Oort	0.281	0.707	0.017 *
GCPS vs Power of Choice	0.281	0.492	0.126
GCPS vs FedAvg	0.281	0.436	0.067
GCPS vs FedProx	0.281	0.377	0.329
GCPS vs MaxVisibility	0.281	0.169	0.241

Paired t-tests with four degrees of freedom. Newly added baselines shown in bold

Fig. 6 Graph Structure Preservation. Spectral preservation ratio for four methods (Proposed, FedAvg, MaxVisibility, GeoDiversity). The comparison focuses on the contrast between structural connectivity (Proposed) and spatial coverage (GeoDiversity) methods. Proposed maintains highest algebraic connectivity of the training subgraph

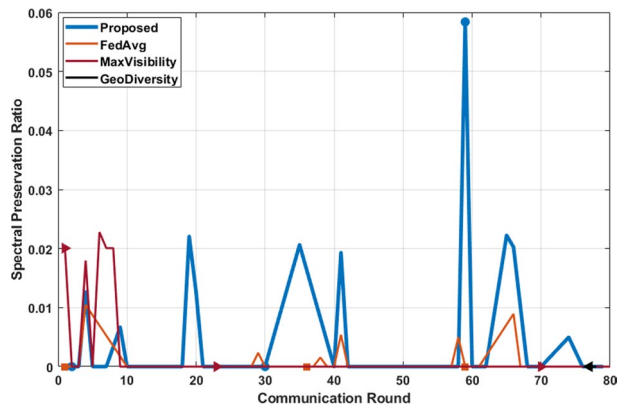
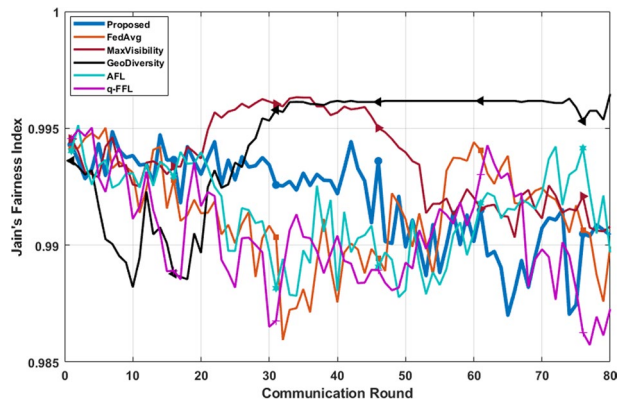


Fig. 7 Per Node Accuracy Fairness. Jain's index for six methods (Proposed, FedAvg, MaxVisibility, GeoDiversity, AFL, q-FFL). Despite all shown methods exceeding 0.98, several produce zero-recall failures (see Figure 5), demonstrating the limitations of aggregate fairness metrics for safety-critical evaluation



6.4 Fairness and Participation Equity

Figure 7 displays Jain's fairness index. All methods achieve high fairness values (exceeding 0.98), yet GeoDiversity, GraphWeighted, Oort, AFL, and q-FFL all produce zero-recall nodes. This demonstrates that Jain's index measures participation equity rather than performance equity. This critical distinction is visualized in Figure 9g.

Figure 8 reveals participation patterns. GCPS produces a broad, bell shaped selection count distribution. MaxVisibility favors high latitude stations. GeoDiversity repeatedly selects a sparse subset while neglecting others.

6.5 Ablation Study

The ablation study expanded to include safety metrics is shown in Figure 9a. The full GCPS configuration with bandit achieves the best balance of safety metrics: FNR of 0.184 and Worst Recall of 0.311. Removing the diversity component causes Worst Recall to collapse to 0.014 (near-complete blindness), demonstrating that the anti-recency mechanism is essential for ensuring all stations participate. Removing the spectral component similarly causes collapse to 0.020, proving that algebraic connectivity preservation is critical. Fixed Weights achieves better accuracy (0.534 versus 0.509) but worse safety (Worst Recall 0.160 versus 0.311, FNR 0.261 versus 0.184), confirming that the bandit adaptation shifts emphasis toward safety metrics.

6.6 Sensitivity Analysis

The sensitivity analysis in Figure 9b-e shows GCPS demonstrates comparable or superior robustness to FedAvg and Oort across all parameters. The topology mismatch sensitivity (0 to 20%) addresses the static graph assumption concern.

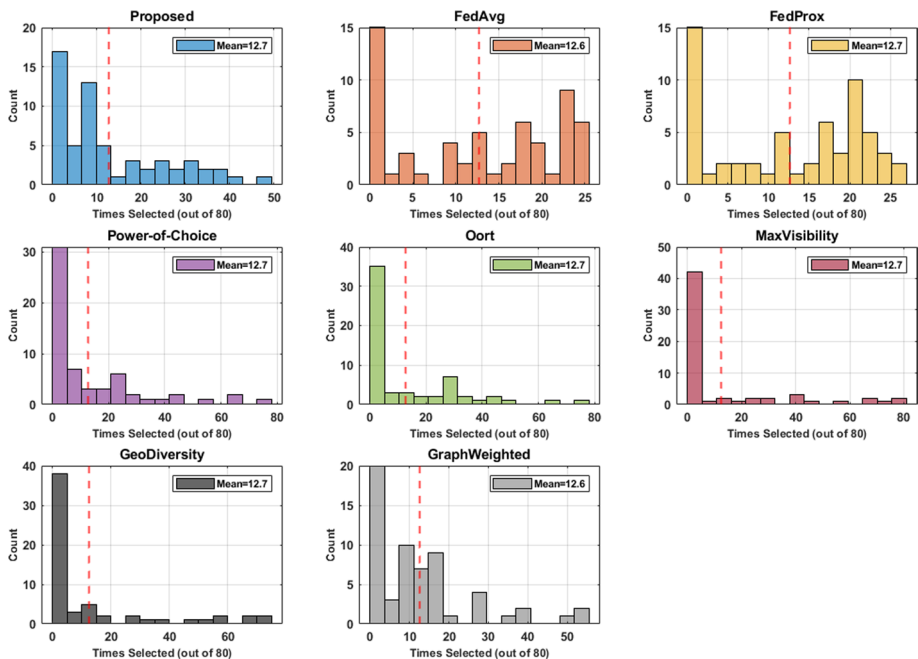


Fig. 8 Node Selection Distribution. Per node selection counts across eighty communication rounds for eight methods. AFL and q-FFL are omitted as they use random selection (distribution identical to FedAvg). Proposed shows most uniform distribution (bell-shaped), while others show skewed or concentrated patterns. Red dashed lines indicate mean selection count

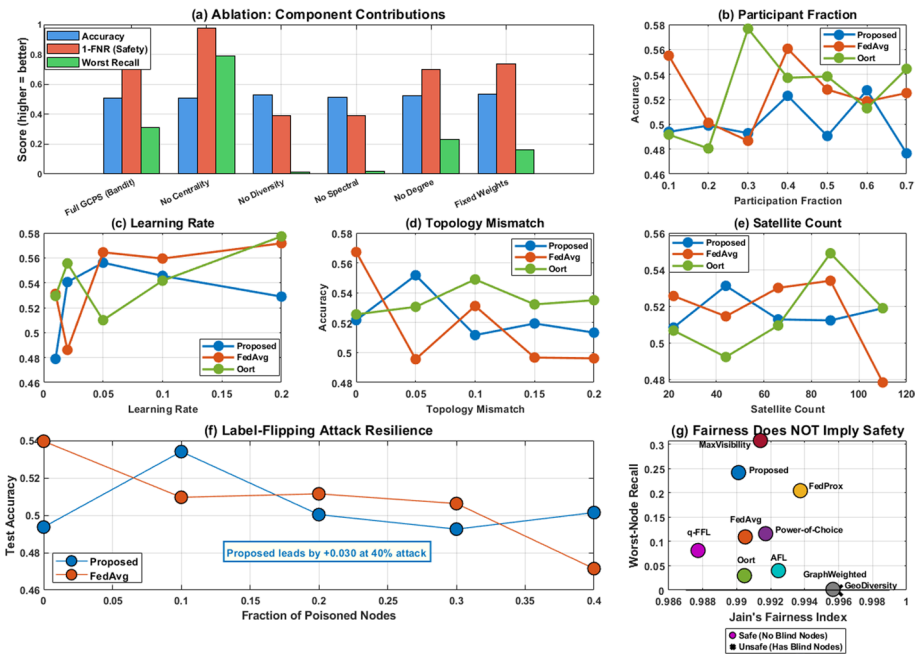


Fig. 9 Comprehensive Analysis. **a** Ablation study comparing six GCPS configurations on accuracy, 1-FNR, and Worst Recall. **b-e** Sensitivity analysis for three methods (Proposed, FedAvg, Oort) across four parameters. **(f)** Label-flipping attack resilience comparing Proposed versus FedAvg. **g** Fairness versus Safety analysis with all 10 methods, showing that high Jain's index does not imply safe deployment. Methods with fairness exceeding 0.99 (GeoDiversity, GraphWeighted) produce blind nodes in 80 to 100% of trials

6.7 Adversarial Robustness

Figure 9f presents resilience to label flipping attacks. At 40% attack strength, GCPS maintains higher accuracy than FedAvg (difference of +0.030), attributable to the diversity term preventing repeated selection of compromised nodes.

6.8 Structural Connectivity Analysis

Figure 10 provides spatial analysis of the selection patterns.

Thus summarizing, The empirical evaluation supports six central claims. First, GCPS substantially reduces zero recall nodes (0 out of 5 trials, 0%) compared to GeoDiversity (5 out of 5, 100%), GraphWeighted (4 out of 5, 80%), AFL (2 out of 5, 40%), and q-FFL (2 out of 5, 40%), visualized directly in Figure 5. Second, GCPS significantly reduces false negative rates compared to GeoDiversity, GraphWeighted, q-FFL, AFL, and Oort (Figure 4, Table 4). The superficially low FNR of MaxVisibility illustrates why multiple safety metrics are essential: its low FNR coexists with the worst minimum per node accuracy, demonstrating that favorable aggregate false negative performance can be achieved by systematically neglecting certain stations. Third, safety improvements are achieved without sacrificing standard accuracy (Figure 1). Fourth, safety guarantees emerge from interaction of all four

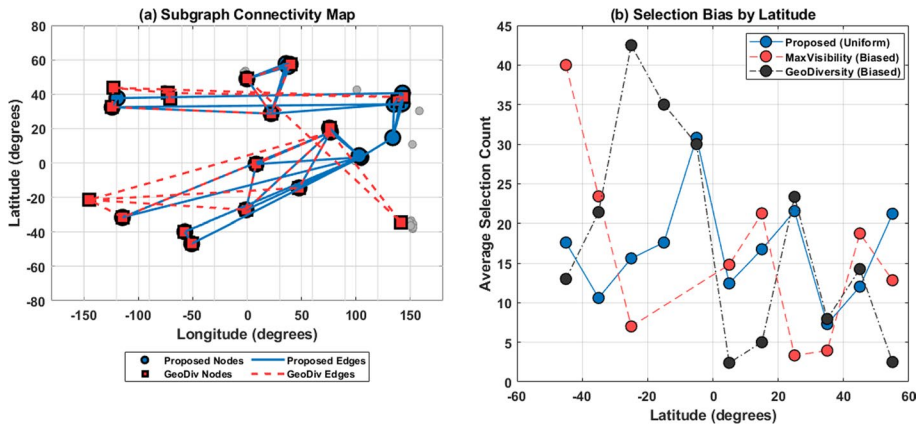


Fig. 10 Structural Analysis. **a** Subgraph Connectivity Map comparing Proposed (blue nodes, solid edges) versus GeoDiversity (red nodes, dashed edges). This focused comparison highlights the central distinction between structural connectivity (Proposed, averaging 44 edges) and spatial coverage (GeoDiversity, averaging 23 edges). **b** Selection Bias by Latitude for three methods (Proposed, MaxVisibility, GeoDiversity), illustrating how MaxVisibility and GeoDiversity exhibit strong latitudinal bias while Proposed maintains uniform coverage

scoring components, with participation patterns (Figure 8) and spectral preservation (Figure 6) providing mechanistic explanation. Fifth, GCPS demonstrates robustness to parameter variation (Figure 9b-e) and adversarial attacks (Figure 9f). Sixth, the fairness versus safety distinction (Figure 9g) establishes that aggregate fairness metrics are insufficient for safety-critical evaluation, as high Jain's index masks catastrophic zero-recall failures.

7 Discussion

The empirical findings carry several implications for practical deployment of federated learning over satellite networks. The most immediate is that communication centric scheduling heuristics, despite their operational prevalence and intuitive appeal, are fundamentally unsuited for safety critical learning tasks. Geographic diversity scheduling, which explicitly aims to ensure broad spatial representation, paradoxically produces the worst safety outcomes. The quantitative analysis reveals the explanation: GeoDiversity selects geographically dispersed but structurally isolated nodes (subgraphs with only 23 edges on average) that cannot effectively share model updates, while GCPS selects connected subgraphs (44 edges on average) enabling information flow. This finding underscores that graph theoretic connectivity, not geographic coverage, determines whether collaborative learning succeeds across heterogeneous regions.

The addition of AFL and q-FFL baselines provides an important insight: loss-based fairness methods, which reweight client contributions to focus on worst-performing nodes, do improve over GeoDiversity (reducing zero-recall from 100% to 40%). However, they fundamentally cannot eliminate zero-recall nodes because they lack structural connectivity awareness. The proposed scheduling-centric solution is therefore complementary to model-

side optimizations. Model-side methods improve what each client learns, while GCPS improves which clients learn together.

The conditions under which GCPS provides the greatest benefit are precisely those that characterize safety critical satellite sensing: non identically distributed labels with minority class events concentrated in specific geographic regions, combined with periodic availability patterns that create systematic participation asymmetries. When all stations share identical label distributions or when availability is uniform, simpler schedulers may suffice. However, these conditions rarely hold in practice. Environmental events exhibit strong spatial structure, and satellite visibility inherently favors higher latitudes in Walker Delta configurations. The proposed method is therefore tailored to the realistic case rather than the idealized one.

The computational overhead of GCPS relative to random selection is modest (approximately 3.3 ms versus 1.7 ms per round for $N = 60$) and is dominated by the one time pre-processing cost of eigendecomposition. For operational deployments, scalability guidance is as follows: full eigendecomposition is acceptable for N at most 100, sparse eigs() for N between 100 and 1000, and power iteration or Nystrom approximation for N exceeding 1000. The safety guarantees provided by GCPS, namely the substantial reduction of zero recall nodes and significant reduction in false negative rates, justify this overhead in any setting where model failure carries human safety consequences. A tsunami warning system that misses all events in one ocean quadrant is not merely less accurate; it is a liability. GCPS ensures that such catastrophic localized failures do not occur.

8 Conclusion

This work introduced Graph Centric Periodic Scheduling, a client selection framework preserving algebraic connectivity of the training subgraph under periodic, spatially correlated availability. Jointly optimizing betweenness centrality, participation diversity, spectral importance, and node degree with bandit adapted weights, the method was evaluated under realistic Walker Delta orbital dynamics with non identically distributed labels across sixty ground stations against nine baselines. The central finding establishes that graph spectral scheduling substantially reduces catastrophic zero recall node failure (0 out of 5 trials, 0% versus GeoDiversity 5 out of 5, 100%), a guarantee no baseline provides. Even state-of-the-art fairness methods AFL and q-FFL produce blind nodes in 40% of trials. GeoDiversity, GraphWeighted, and Oort each produced ground stations with zero recall across independent trials, rendering entire geographic regions blind to safety critical events. The proposed method maintained non zero recall for every station in every trial while reducing false negative rates by factors exceeding three compared to GeoDiversity and GraphWeighted, both statistically significant ($p < 0.01$). Safety improvements were achieved without sacrificing aggregate accuracy. Beyond the algorithmic contribution, this work argues that minimum per node performance guarantees must become primary evaluation criteria whenever safety critical decisions depend on model outputs, as aggregate metrics conceal catastrophic localized failures invisible under standard protocols. It is further established that commonly used fairness indices do not capture performance equity, and safety-critical evaluation requires disaggregated per-node metrics. The proposed graph spectral approach establishes a prin-

cipled connection between algebraic graph theory and federated learning safety not previously explored.

9 Limitations and Future Work

Several limitations define directions for future investigation. First, the satellite visibility model uses circular orbits with simplified positioning; future work should incorporate SGP4 propagation with real two line element sets from operational constellations. Second, the static communication graph assumption should be extended to dynamic topologies with time varying link qualities, using approximate spectral methods (power iteration, Nystrom sampling) to maintain computational efficiency. Third, formal differential privacy guarantees via moments accountant or Renyi DP frameworks remain absent; only empirical privacy-utility tradeoffs are currently demonstrated. Fourth, spectral scores require cubic eigendecomposition; while acceptable for N at most 100, approximate methods are needed for mega constellations, and their impact on safety guarantees requires further study. Fifth, evaluation remains simulation based, lacking validation against real satellite ground contact traces or hardware testbeds. Sixth, safety metrics are limited to recall and false negative rate; calibration error, out of distribution detection, and covariate shift robustness merit inclusion. Seventh, the discrete seven arm bandit limits adaptation granularity; continuous optimization via Bayesian methods may improve responsiveness. Eighth, theoretical convergence guarantees linking algebraic connectivity to federated learning convergence rates remain an open question. These limitations do not diminish the central contribution: demonstrating that graph spectral scheduling provides minimum per node performance guarantees existing methods cannot match, and establishing that such guarantees must become standard evaluation criteria for safety critical federated learning over intermittent satellite networks.

Author Contributions I am the only author.

Data Availability No datasets were generated or analysed during the current study.

References

- Abtahi, N., Chaman, U. M., & Kabir, M. S. (2026). Federated learning in healthcare: A comprehensive survey on privacy, scalability and clinical applications. *ICT Express*. <https://doi.org/10.1016/j.ict.2026.05.011>
- Choi, Chang-Sik & Ku, Bon-Jun & Baccelli, Francois. (2024). Stochastic geometry and dynamical system analysis of walker constellation networks. <https://doi.org/10.48550/arXiv.2412.01610>.
- de Curtò, J., de Zarzà, I., Roig, G., Cano, J.-C., Manzoni, P., & Calafate, C. (2023). LLM-Informed Multi-Armed Bandit Strategies for Non-Stationary Environments. *Electronics*, 12, 2814. <https://doi.org/10.3390/electronics12132814>
- Dobariya, R., & Hobiger, T. (2026). Automatic scheduling of satellite tracking tasks by means of policy-gradient reinforcement learning and transformer-based pointer networks. *CEAS Space Journal*. <https://doi.org/10.1007/s12567-026-00726-y>
- Fu, L., Zhang, H., Gao, G., Zhang, M., & Liu, X. (2023). Client selection in federated learning: Principles, challenges, and opportunities. *IEEE Internet of Things Journal*. <https://doi.org/10.1109/JIOT.2023.3299573>
- Ji, B., Han, J., Kim, D., Yun, J., & Joo, C. (2026). Inter-satellite link technologies and applications in Low Earth Orbit satellite networks. *ICT Express*, 12(3), 576–592. <https://doi.org/10.1016/j.ict.2026.03.001>

- Ji, X., Zhang, Y., Zheng, K., et al. (2025). Federated asynchronous graph attention network with structural semantic embedding for multi-label graph classification. *Scientific Reports*, 15, Article 38618. <https://doi.org/10.1038/s41598-025-22500-6>
- Johnson, Dr. (2024). Wireless sensor networks for earthquake and natural disaster detection. *American Journal of Sensor Networks and Wireless Communications*, 5, 1–12. <https://doi.org/10.71465/ajsnwc.3193>
- Kim, D., Woo, H., & Lee, Y. (2024). Addressing bias and fairness using fair federated learning: A systematic literature review. <https://doi.org/10.20944/preprints202410.1060.v1>.
- Lagunas, E., & Ottersten, B. (2024). Low-earth orbit satellite constellations for global communication network connectivity. *Nature Reviews Electrical Engineering*. <https://doi.org/10.1038/s44287-024-00088-9>
- Li, Y., Liu, L., Li, H., Liu, W., Chen, Y., Zhao, W., Wu, J., Wu, Q., Liu, J., and Lai, Z. (2024). Stable Hierarchical Routing for Operational LEO Networks. In Proceedings of ACM Conference, pp. 296–311. <https://doi.org/10.1145/3636534.3649362>.
- Li, Y., Zhang, C., Jia, C., Li, X., & Zhu, Y. (2019). Joint optimization of workforce scheduling and routing for restoring a disrupted critical infrastructure. *Reliability Engineering & System Safety*, 191, Article 106551. <https://doi.org/10.1016/j.res.2019.106551>
- Mateo-Garcia, G., Veitch-Michaelis, J., Purcell, C., Longépé, N., Reid, S., Anlind, A., Bruhn, F., Parr, J., & Mathieu, P. (2023). In-orbit demonstration of a re-trainable machine learning payload for processing optical imagery. *Scientific Reports*. <https://doi.org/10.1038/s41598-023-34436-w>
- Paterson, C., Hawkins, R., Picardi, C., Jia, Y., Calinescu, R., & Habli, I. (2025). Safety assurance of Machine Learning for autonomous systems. *Reliability Engineering & System Safety*, 264, Article 111311. <https://doi.org/10.1016/j.res.2025.111311>
- Pavel, M. I., Hu, S., Pratama, M., & Kowalczyk, R. (2026). Onboard Optimization and Learning: A Survey. *IEEE Access*, 14, 33072–33104. <https://doi.org/10.1109/ACCESS.2026.3664956>
- Pelton, J.N., Laufer, R. (2020). Commercial Small Satellites for Business Constellations Including Micro-satellites and Minisatellites. In: Pelton, J.N., Madry, S. (eds) Handbook of Small Satellites. Springer, Cham. <https://doi.org/10.1007/978-3-030-36308-6>
- Satish, S., Gonaygunta, H., Yadulla, A. R., Kumar, D., Maturi, M. H., Meduri, K., De La Cruz, E., Nadella, G. S., & Sajja, G. S. (2025). Forecasting the unseen: Enhancing tsunami occurrence predictions with machine-learning-driven analytics. *Computers*, 14(5), Article 175. <https://doi.org/10.3390/computers14050175>
- Shinde, S. S., Naseh, D., Tarchi, D., & Fischione, C. (2026). LEO satellites accelerating edge intelligence: Model transfer and similarity-aware initialization for federated learning. *IEEE Internet of Things Magazine*. <https://doi.org/10.1109/MIOT.2026.3700215>
- Wang, B., Li, A., Li, H. H., & Chen, Y. (2020). GraphFL: A Federated Learning Framework for Semi-Supervised Node Classification on Graphs. *IEEE International Conference on Data Mining (ICDM)*, 2022, 498–507. <https://doi.org/10.1109/ICDM54844.2022.00060>
- Waref, D., Alayary, Y., & Abd El Ghany, M. A. (2026). Review on federated optimization in multi-tier architectures. *Journal of King Saud University - Computer and Information Sciences*, 38, Article 101. <https://doi.org/10.1007/s44443-026-00510-2>
- Zaidyn, M., Akhtanov, S., Turlykozhaeva, D., Temesheva, S., Akhmetali, A., Skabylov, A., & Ussipov, N. (2025). The role of fractal dimension in wireless mesh network performance. *Scientific Reports*, 15(1), 42149. <https://doi.org/10.48550/arXiv.2506.19366>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Authors and Affiliations

Bilal Ahmad¹ 

✉ Bilal Ahmad
bahmad@metu.edu.tr

- ¹ Electrical and Electronics Engineering, Middle East Technical University-NCC, Kalkanlı, Güzelyurt, 10, 99738 Mersin, Turkey